

E-commerce players and food delivery services are trying to increase speed of delivery from same-day to under 30 mins and now even under 10 mins. This reduction in delivery time wouldn't have been possible without increased processing speed of the request inside a data centre somewhere, I can guarantee you that.

d Challenges of setting up and operating a data centre in India?

Sunil Gupta: Behind the obvious digitisation layer, there is an underlying infrastructure layer dealing with land procurement, getting multiple approvals and clearances, getting lots of power, storing a huge amount of fuel, getting environmental clearances, etc. Once these challenges are overcome, then we arrive at the data centre constructors and operators from an IT standpoint. The shortage of really high-skilled people, those who really understand the designing, engineering and operations of the data centre is very real. Having spent over 20 years in this business I can literally count the number of these highly-skilled people on my fingertips – and they won't go into double digits, believe me. For an industry of this scale, which is supposed to become seven times bigger in the next five years, it's a scary scenario.

One way we are trying to solve this high-skilled labour shortage facing the data centre sector in India is by hiring people from allied industries. So somebody may be working in the chemical industry or in a nuclear plant or having experience in sophisticated industry work, we actively hire them. We also have a program of taking on fresh graduates in a technical course and training them on the job here at Yotta.

d Why is it necessary to have more data centres within India?

Sunil Gupta: A whole lot of innovation will happen at the mobile device level, at the sensor level, at the smart machine level, and there are a whole lot of startups and good companies who are doing work in this space.

YOTTA NM1: IN NUMBERS...

The Yotta NM1 Datacentre has a power plant that can power 50,000 families, enough fuel to fill up 11,000 SUVs, 400 metric tonnes of cooling, enough CCTVs for 10 high-security facilities and one expensive piece of paper which certifies it as Asia's largest Tier-IV datacentre. It's almost impenetrable with a state-of-the-art seven layers of security cover enabled by technologies like: Under Vehicle Scanning System, Automatic Number Plate Recognition System, 1200 CCTVs and AI monitoring every inch of the facility and its perimeter.



In the heart of the datacentre, there are six dedicated server hall floors, each can house 1200 racks or up to 52-U with 1500 kg of load per sq metre. There are 7200 server racks in just one building inside the Yotta NM1, while an additional 30,000 racks are coming up soon in adjoining structures inside the datacentre park. The terrace has an MS-Grille platform with 34 chillers of 400 TR to serve as chilled water distribution plants. Hot water coming out of the server racks is usually at 28 degrees, which is chilled down to 15 degrees and recirculated for cooling the servers. And as Sunil Gupta himself confirmed, there's roughly 1 Tbps of data flowing in and out of the datacentre every single second, and the entire facility's electricity consumption is in the range of crores of rupees per month.

Very soon 5G is going to get started in India, where all the telecom operators will compete to provide great network connectivity across the country. As a result, data generated by Indians is expected to go through the roof over the next five years!

Hence, India needs to focus on the IT layer on the top of the datacentre which is represented by cloud, which is primarily hosted by Amazon's AWS, Microsoft Azure and Google's Cloud. There's a need to have more Indian cloud operators to offer Indian businesses more choice. We also have to be sensitive towards geopolitical situations around the world right now, where we see countries get-

ting isolated because of their foreign policy or bordering countries – for example, the situation between India and China. Some of the Indian data which the Indian government feels is absolutely critical needs to be in Indian hands, comply with Indian law, and should be processed and stored here in India and nowhere else. But doing so is easier said than done, because companies like Amazon and Google spend billions of dollars, building so much intelligence and core competency in this sector, but we definitely have to start somewhere and up our R&D spend to innovate around these gaps in the roadmap as efficiently as we can. **d**